An Introduction to Google Machine Learning APIs

This article gives readers an overview of Google machine learning APIs. It contains code samples to get started with APIs, bringing the power of machine learning to everyday developers.

For the past two decades, Google has been at the forefront of collecting and mining data. It has built multiple machine learning models through which it now provides some of the most powerful machine learning APIs. These machine learning models are the magic behind some of the applications like Google Photos, Inbox Reply (Gmail), Google Translate and more.

In the last year, Google has opened up its machine learning models by making its APIs accessible to the public. The Google Cloud Machine Learning Platform provides easy-to-use REST APIs that have powerful capabilities like computer vision, NLP, translation and more.

Google Cloud Machine Learning Platform

One of the main issues facing developers is figuring out how much of theoretical knowledge they need before using machine learning algorithms in their applications. The team at Google has understood this and identified a spectrum across which different products from Google can be used from the Machine Learning Platform.

At one end of the spectrum is TensorFlow, which is an open source library for machine intelligence. TensorFlow does require that you are up to speed on various machine learning concepts and can use the API to build machine models, using them effectively in your applications. This might not be possible for all developers, especially those who are looking for REST APIs to use in their applications. To cater to this need, Google has identified multiple capabilities in its Machine Learning Platform and has exposed those capabilities in the form of easy-to-call REST APIs. There is also a free quota per month for these APIs.

Over the last few months, Google has released multiple REST APIs in the Machine Learning Platform. The most important ones are listed below.

Google Cloud Vision API: The Cloud Vision API provides a REST API to understand and extract information from an image. It uses powerful machine learning models to classify images into thousands of categories, detect faces, identify adult content, emotions, OCR support and more. If you are looking to classify and search through images for your application, this is a good library to consider.

Google Natural Language API: The Natural Language API is used to identify parts of speech and to detect multiple types of entities like persons, monuments, etc. It can also perform sentiment analysis. It currently supports three languages: English, Spanish and Japanese.

Google Speech API: The Speech API is used to translate audio files into text. It is able to identify over 80 languages and their variants, and can work with most audio files.

There are multiple other APIs in this family like the Translate API, but we will limit our sample code to the above three APIs.

Getting started

To get started with using the machine learning APIs listed above, you need to sign up for the Google Cloud Platform and